

Lab 3: The Linear Algebra of Least Squares

PSTAT 134 (Winter 2026)

Reference: Calder & Olver, Linear Algebra, Data Science, and Machine Learning (Springer, 2025)

Question 1: Deriving the Hat Matrix In the general linear regression model $y = X\beta + \varepsilon$, the method of Ordinary Least Squares (OLS) seeks the vector $\hat{\beta}$ that minimizes the sum of squared residuals, $RSS = (y - X\beta)^\top (y - X\beta)$. The solution to this minimization is the vector $\hat{\beta}$ that satisfies the normal equations $(X^\top X) \hat{\beta} = X^\top y$.

- Assuming X has full column rank, solve for $\hat{\beta}$ and substitute it into the fitted values equation $\hat{y} = X\hat{\beta}$. Show that $\hat{y} = Hy$, where $H = X(X^\top X)^{-1}X^\top$. (This is why H is called the “Hat Matrix”—it puts the “hat” on y). *[Hint: Use Section 3.6 on matrix inverses and Section 4.7.2 on least squares.]*
- Show that H is a symmetric matrix. *[Hint: Use Definition 3.1 (symmetric matrices), Lemma 3.2, and Proposition 3.27 on transposes of products.]*
- Show that H is an idempotent matrix (i.e., $H^2 = H$). *[Hint: See Section 4.6 on projection matrices.]*

Question 2: The Geometry of Projection A fundamental result in linear algebra is that for any subspace V , a vector y can be uniquely decomposed as $y = u + v$, where $u \in V$ and $v \in V^\perp$ (Section 2.6). The vector u is the orthogonal projection of y onto V . Using the Hat Matrix H defined in Question 1, let the vector of fitted values be $\hat{y} = Hy$ and the vector of residuals be $e = y - \hat{y} = (I - H)y$.

- Prove that the fitted values $\hat{y}$ lie within the image of X , denoted $\text{img}(X)$. *[Hint: Use Definition 3.6 and equation (3.26) from Section 3.4.]*
- Prove that the residuals e are orthogonal to every column in X (i.e., show that $X^\top e = 0$). *[Hint: This relates to the Fundamental Theorem of Linear Algebra (Theorem 4.24) and equation (4.56).]*

- c. Based on (a) and (b), conclude that Hy is the unique orthogonal projection of the data vector y onto $\text{img}(X)$. [Hint: Use Section 2.4 on orthogonal projection and Section 4.6 on projection matrices.]

Question 3: Eigenvalues of Idempotent Matrices Now consider the general algebraic properties of the matrix H identified in Question 1. Let A be any $n \times n$ matrix that is idempotent ($A^2 = A$).

- a. Let λ be an eigenvalue of A with corresponding eigenvector x . Prove that λ must be either 0 or 1. [Hint: Use the definition of eigenvalues from Section 5.1.]
- b. Using the result from (a) and the Spectral Theorem (Theorem 5.22 in Section 5.3.1, which states that symmetric matrices have real eigenvalues and an orthonormal eigenvector basis), explain why a symmetric idempotent matrix is positive semidefinite. [Hint: Use Definition 4.8 and Theorem 4.12 on positive semidefinite matrices from Section 4.1.]

Question 4: Rank, Trace, and Degrees of Freedom In statistical hypothesis testing, the distribution of quadratic forms (sums of squares) often depends on the rank of the matrix involved.

- a. Prove that for any idempotent matrix A , the rank of A equals the trace of A ($\text{rank}(A) = \text{tr}(A)$). [Hint: Use the Spectral Theorem and the fact that trace equals the sum of eigenvalues. See Exercise 1.15 for trace properties.]
- b. Calculate the trace of the Hat Matrix H defined in Question 1. [Hint: Use the cyclic property of trace from Exercise 1.15(c): $\text{tr}(ABC) = \text{tr}(BCA)$.]
- c. Combine these facts to state the degrees of freedom associated with the projection H (and thus the model variance) in a full rank regression model. [Hint: Relate to Definition 3.6 on rank.]

Question 5: Leverage and Diagonal Entries of the Hat Matrix The diagonal elements of the Hat Matrix H , denoted by h_{ii} , are called **leverages** and play an important role in regression diagnostics.

- a. Show that for each $i = 1, \dots, n$,

$$h_{ii} = x_i^\top (X^\top X)^{-1} x_i,$$

where $x_i^\top$ is the i -th row of the design matrix X . [Hint: Use the matrix multiplication formula (3.10) and the definition of H .]

- b. Prove that

$$0 \leq h_{ii} \leq 1 \quad \text{for all } i.$$

[Hint: For the lower bound, use Theorem 4.13 on Gram matrices. For the upper bound, use the idempotent property from Question 1(c).]

c. Show that

$$\sum_{i=1}^n h_{ii} = p,$$

and explain how this result relates to the degrees of freedom of the regression model.
[Hint: Use Question 4(b) and Exercise 1.15 on trace.]